

KARAN CHAWLA DORA

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EDUCATION

University of Toronto

Bachelor of Applied Science — Engineering Science (cGPA: **3.95/4.0**)

Toronto, ON

2025 – 2030

Research Interests: control systems, robot autonomy, state estimation

EXPERIENCE

aUToronto — SAE AutoDrive

Toronto, ON

3D Mapping Lead (promoted)

Jul. 2026 – Present

- Lead a team of two building SLAM pipelines; generated HD LiDAR maps of UTIAS by configuring and tuning LIO-SAM, integrating GPS via navsat, and converting raw IMU data to physical units.
- Develop dynamic-object removal workflows and extend mapping to public-road ROS bag recordings.

State Estimation Developer

Sep. 2025 – Jul. 2026

- Contributed Python validation scripts to sweep Iterated Extended Kalman Filter (IEKF) noise parameters and lane-update settings, comparing trajectories against ground-truth GPS under GPS attenuation.
- Tested LiDAR localization pipelines on ROS 2 bags, contributing to the team's **1st-place localization and overall finishes** at SAE AutoDrive Challenge R2Y5.

FIRST Robotics Competition

Jul. 2021 – Jun. 2025

Robotics Team Captain & Software Development Lead

Ottawa, ON

- Led a 90+ member team through design, manufacturing, and software development; the team qualified for the World Championship and earned two Provincial Finalist finishes.
- Built C++ autonomous routines using AprilTag vision, wheel odometry, and trajectory generation; team average autonomous points rose approximately 3x from the first competition to provincials.
- Tuned feedforward and PID feedback controllers to improve robot mechanism precision.

RESEARCH EXPERIENCE

Autonomous Space Robotics Lab (ASRL)

May 2026 – Present

Robotics Researcher — Supervisor: Tim Barfoot

UTIAS

- Deployed Doodle Labs mesh radios on four AgileX Hunter robots and a base station using OpenWrt and custom firewall rules, replacing centralized Wi-Fi with a decentralized B.A.T.M.A.N. network.
- Verified 400+ m line-of-sight connectivity, non-line-of-sight operation, and recovery from node dropouts through field testing, overcoming the prior network's 5 m robot-separation limit.
- Built a ROS 2 node publishing network telemetry and battery health for planned dataset collection.

Bruce Lab — University of Ottawa | Research Assistant

Aug. 2024

- Implemented C++ PID controllers for platinum-nickel electrodeposition, reducing transient time by approximately 80% and steady-state error by approximately 40% versus the baseline.
- Automated experiment execution, sensor logging, and data curation with a Python GUI, replacing manual component operation and reducing experiment completion time by approximately 33%.
- Refined an Arduino-based potentiostat circuit and control workflow, expanding the operating voltage range by approximately 5x.

PROJECTS

Q-Learning Trainer | Python, FastAPI, NumPy, JavaScript

June 2026

- Built a tabular Q-learning simulator with custom maps and real-time WebSocket visualization, enabling interactive training, hyperparameter experimentation, and model evaluation.
- Deployed the dashboard to Fly.io using Docker and session isolation; added pytest regression tests for map parsing, reachability, and policy evaluation.

TECHNICAL SKILLS

Languages: Python, C++, C, Bash, Java, JavaScript, TypeScript

Robotics & Control: ROS 2, SLAM, LIO-SAM, LiDAR mapping, state estimation, sensor fusion, PID, AprilTags

Tools & Systems: Linux, Git, Docker, OpenWrt, mesh networking, Arduino, NumPy, pandas, pytest